JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 94, NO. B12, PAGES 17,881-17,888, DECEMBER 10, 1989

PALEOMAGNETISM OF THE DEWEY LAKE FORMATION (LATE PERMIAN), NORTHWEST TEXAS: END OF THE KIAMAN SUPERCHRON IN NORTH AMERICA

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Abstract. The remanent magnetization of red beds of the Late Permian Dewey Lake Formation from northwest Texas was investigated using thermal, chemical, and alternating field demagnetization. After removal of a small present-day viscous overprint, the Dewey Lake shows univectorial decay of the remanence revealing a dual-polarity characteristic direction of D=152.4°, I=-11.1°, k= 39.5, α_{95}=5.9°, N=16 sites. Three polarity transitions were observed defining at least four polarity zones. The observed polarity zones are roughly layer parallel and may correlate with the Illawarra magnetozone at the end of the Kiaman. If this is so, K-Ar age determinations of an interbedded ash layer (Kolker and Fracasso, 1985) suggest that the Kiaman Superchron ended before 251±4 Ma. The paleopole, located at 51.4°N-126.2°E, confirms the reliability of previously reported Late Permian poles from southwest United States, which fall significantly off the Late Permian segment of the paleomagnetic Euler pole (PEP) track of Gordon et al. (1984). This implies that a more segmented set of small circles is required to model the North American apparent polar wander path, diminishing the usefulness of the PEP method.

Introduction

The reverse polarity Kiaman Superchron is perhaps one of the most remarkable features of the geomagnetic field in the past. For a period of about 50 m.y., including most of the Permian and part of the Late Carboniferous, the field maintained reverse polarity. As a global phenomenon, it offers an excellent opportunity for intercontinental chronostratigraphic correlations.

Determination of both the onset and end of the Kiaman Superchron, and therefore its duration, remains, however, unsatisfactory [Opdyke, 1986]. This is due partially to problems in dating continental red bed sequences with normal polarity interpreted to be pre- or post-Kiaman, when these rocks are unfossiliferous and only dated by lithological correlations. The onset of the Kiaman is stratigraphically constrained to the early Westphalian (Early Pennsylvanian) [Roy and Morris, 1983; Opdyke, 1986]. In North America it is known to postdate deposition of the uppermost Mississippian Mauch Chunk Formation in Pennsylvania [Kent and Opdyke, 1985] and to predate deposition of the Minturn Formation in Colorado [Opdyke, 1986], the lower part of which is considered to be Atokan (Middle Pennsylvanian) in age.

The end of the Kiaman Superchron occurs in the Permian, where it is followed by the Illawarra mixed polarity interval of Irving and Parry [1963]. Rarely observed, normal geomagnetic polarities have been reported in Upper Permian sequences in the Soviet Union [Khramov and Rodinov, 1981], northwest Argentina [Valencio et al., 1977], southwest United States [Peterson and Nairn, 1971], and southern China [Zhao and Coe, 1987; Heller et al., 1988]. Thus the upper boundary is placed in the Late Permian.

We report dual-polarity paleomagnetic data from well-dated Upper Permian red beds from stable North America that yield a reliable Late Permian cratonic paleomagnetic pole. This result allows us to constrain the age of the end of the Kiaman interval, and improves the determination of the Late Permian segment of the North American apparent polar wander path (APWP). The paleopole deviates from the paleomagnetic Euler pole (PEP) track of Gordon et al. [1984], suggesting that North American plate motion was not constant relative to the rotation axis throughout the Late Carboniferous-Early Triassic interval, as has previously been argued [Gordon et al., 1984].

Geologic Setting and Sampling

The Dewey Lake Formation, in the Palo Duro Basin, NW Texas (Figure 1) is a flat lying sequence of continental sediments deposited in fluvial and lacustrine environments. It consists principally of interbedded fine-grained red sandstone and siltstone beds and multicolored shales.

The Dewey Lake Formation is unfossiliferous and conformably overlies Upper Permian gypsum, shale, and dolomite of the Alibates Formation. The upper contact with the Lower Triassic Dockum Formation has been described both as conformable and disconformable. Two thin interbedded ash

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Paper number 89JB01504.
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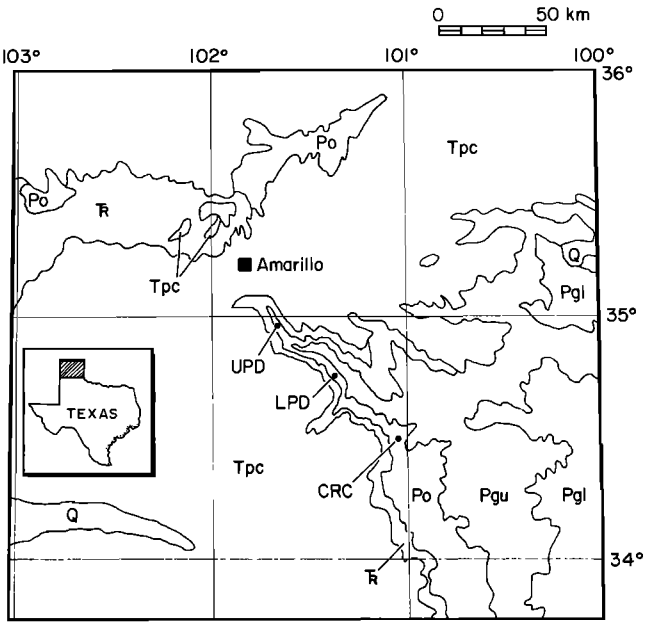


Fig. 1. Generalized geologic map of the Texas Panhandle area showing major stratigraphic elements (Tpc, Tertiary; T, Triassic; Po, Ochoan; Pgl, Guadalupian lower; and Pgu, Guadalupian upper) and sampling localities: UPD, Palo Duro Canyon Texas State Park; LPD, Lower Palo Duro Canyon crossing Texas Highway 207; CRC, Caprock Canyon Texas State Park.

layers were described in the Dewey Lake Formation by Kolker and Fracasso [1985]. Biotite concentrates from the lowermost ash layer produced K-Ar ages ranging between 251±4 and 261±9 Ma [Kolker and Fracasso, 1985]. Split sample determinations in different laboratories yield similar results: Ages of 257 and 261 where obtained at Krueger Geochron Laboratories (Cambridge, Massachusetts) and a younger age of 251 was obtained at the University of Texas at Austin laboratory. Biotite grains are euhedral, suggesting minimal transport or absence of a detrital component. According to the time scale of Harland et al. [1982], this age range corresponds to the Late Permian.

A total of 78 block samples and 22 field-drilled samples (19 sites from individual beds of a few centimeters to 1m in thickness) were collected and oriented with a magnetic compass at three localities: Palo Duro Canyon (UPD) Texas State Park, Caprock Canyon (CRC) Texas State Park, and along the crossing of Texas Highway 207 and the lower Palo Duro Canyon (LPD) (Figure 1). One to four 2.2-cm-high and 2.5-cm-diameter specimens were obtained from each sample.

Paleomagnetic Results and Analysis

Natural remanent magnetizations (NRM) were measured using a two-axis ScT cryogenic magnetometer at the paleomagnetic laboratory of the University of Michigan. NRM intensities are typically of the order of 10^{-2} A/m, ranging between 0.6 and 39.2 mA/m. NRM directions formed two well-grouped clusters with shallow inclinations and both SSE and NNW declinations. Only a few samples showed intermediate directions, commonly from sites showing larger dispersion and lower NRM intensities.

The remanent magnetization was analyzed with standard demagnetization techniques. Pilot samples were subject to stepwise thermal demagnetization at 150°, 50°, 20°, and 10° steps (as temperature increased) up to 700°C. One or two specimens from each sample were then treated with at least six steps. AF demagnetization was performed with Schonstedt equipment, up to a peak induction field of 100 mT. Ten specimens were chemically demagnetized in an ambient field using concentrated HCl at 60°C; samples were then rinsed and dried in a reduced induction space (less than 10 nT) before measurement.

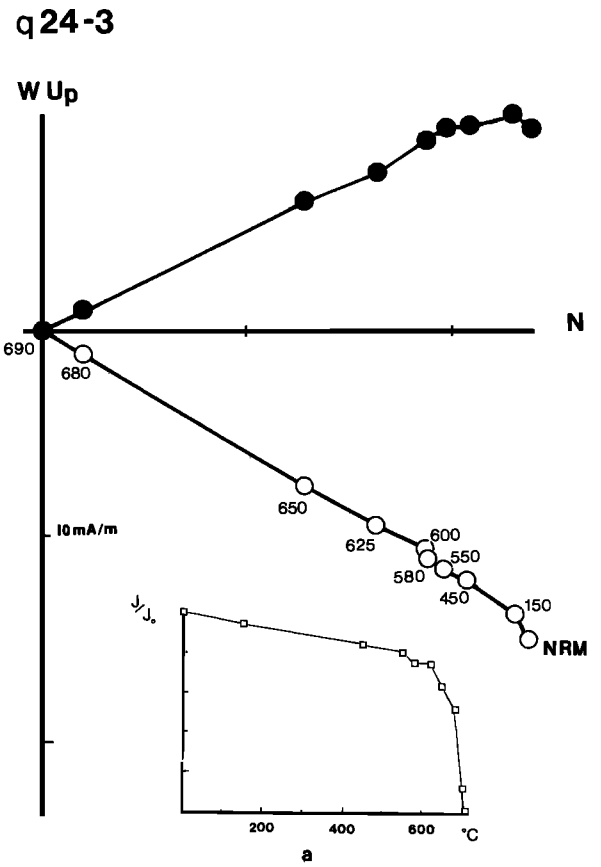
Demagnetization behavior is relatively simple. One or at most two components of magnetization can be distinguished in most samples using orthogonal demagnetization diagrams [Zijderveld, 1967]. Typical demagnetization diagrams (Figure 2) show removal of a small, northerly, and steep direction with laboratory unblocking temperatures of less than 450°C. The remaining component, hereafter called the characteristic direction, is stable up to temperatures greater than 670°C. This component is observed as linear segments trending to the origin in demagnetization diagrams (Figure 2).

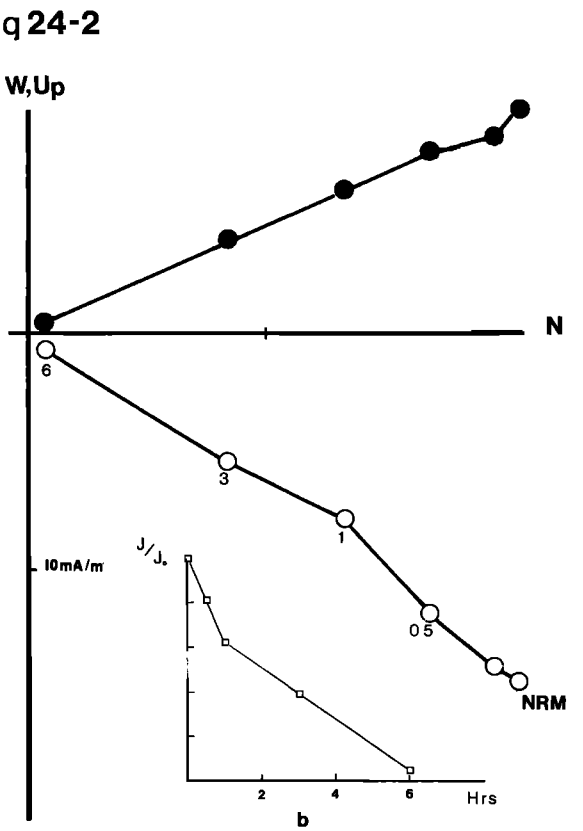
Chemical demagnetization is very effective, partially due to high permeability, although it was not possible to apply this method to all the material collected because most samples contained calcareous cement. Both chemical and thermal demagnetization techniques isolate an identical characteristic direction in specimens of the same sample (Figures 2a and 2b). Normalized intensity decay diagrams for chemical demagnetization show a break in slope when most of the red pigment is dissolved. Microscopic observations reveal that large (approximately 20 μm) hematite grains (as replacement of magnetite) are progressively removed during further leaching. The univectorial behavior, combined with these microscopic observations, implies that both the pigment and the larger hematite grains are carriers of this well-defined remanent magnetization.

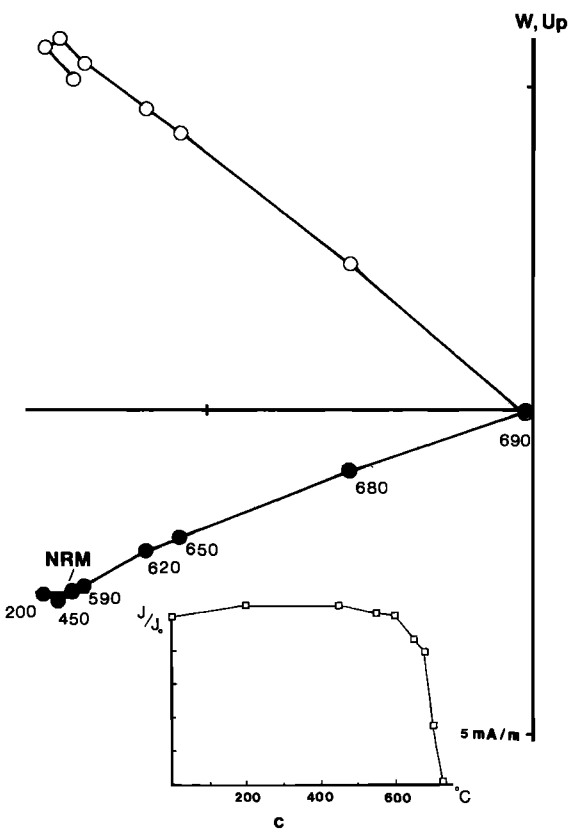
Demagnetization diagrams for samples demagnetized first in an alternating field and then treated thermally (Figure 2d) indicate that AF demagnetization removes only the low

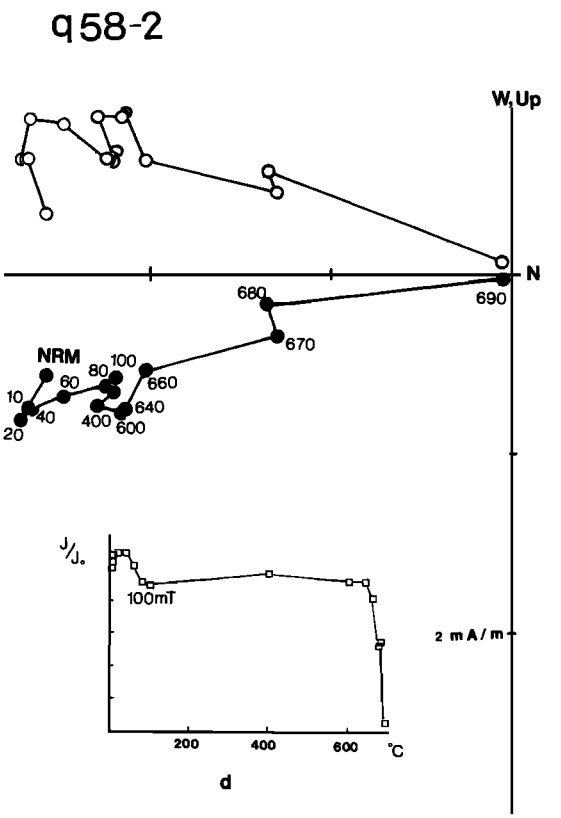
Figure 2. Representative demagnetization diagrams [Zijderveld, 1967]. Open (solid) symbols represent projections onto a vertical (horizontal) plane: (a) and (c) thermal demagnetization, (b) chemical demagnetization, (d) combined AF and thermal demagnetization. Inset on each diagram shows normalized total intensity decay diagrams.

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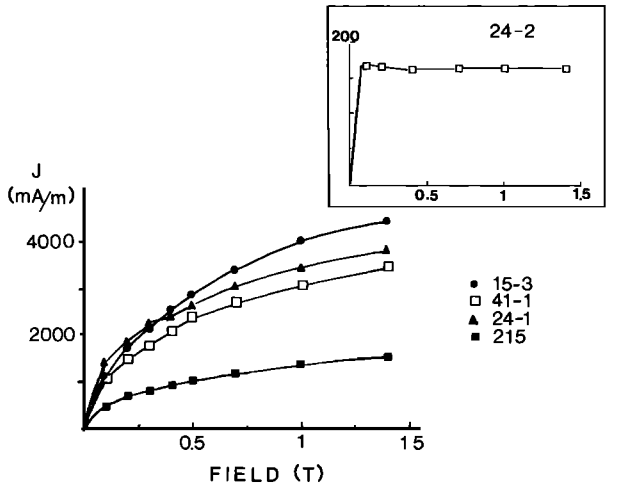


Fig. 3. Representative IRM acquisition curves. The sample in the figure of the inset (sample 24-2) was previously chemically demagnetized.

laboratory unblocking temperature component. The magnetization remaining after 100mT AF treatment is similar in direction to that isolated by thermal and chemical demagnetization methods alone.

The characteristic directions reveal both reverse and normal polarities (i.e., NNW shallow-down and SSE shallow-up). The high unblocking temperatures and coercivities observed indicate that hematite is the essential carrier of the characteristic magnetization. This is further supported by isothermal remanent magnetization (IRM) acquisition experiments (Figure 3) which show curves typical of hematite (+subordinate magnetite). Chemically demagnetized samples show IRM behavior typical of magnetite bearing samples (Figure 3 inset), as expected because magnetite is less soluble than hematite. After chemical treatment, only a very small percentage of the initial NRM (less than 1%) was left in the samples (Figure 2b). This implies that the small fraction of magnetite present in some samples is not contributing significantly to the stable remanence.

Specimen directions were determined by principal component analysis [Kirschvink, 1980], using three or more points in the demagnetization diagram to determine individual directions. The great majority of the samples have maximum angular deviation (MAD) values of less than 2. Only samples with MAD values smaller than 10 were included in the site mean and formation mean calculations. Figure 4a shows the directions obtained for the first-removed component from samples where it is well defined. The mean direction ($D=357.8^{\circ}$, $I=55.3^{\circ}$; $\alpha_{95}=4^{\circ}$, $k=36.7$, $N=36$) is statistically identical to the local present-day dipole field ($I=55^{\circ}$). We interpret this direction as a recent overprint of probable viscous origin.

The characteristic directions (Figure 4b) were observed in all but three sites. Site 1 was from dolomites of the Alibates Formation, where remanent magnetization was weak and no meaningful magnetization could be resolved. Site 4 was located near a transition from reversed to normal polarity and had the lowest NRM intensity of all

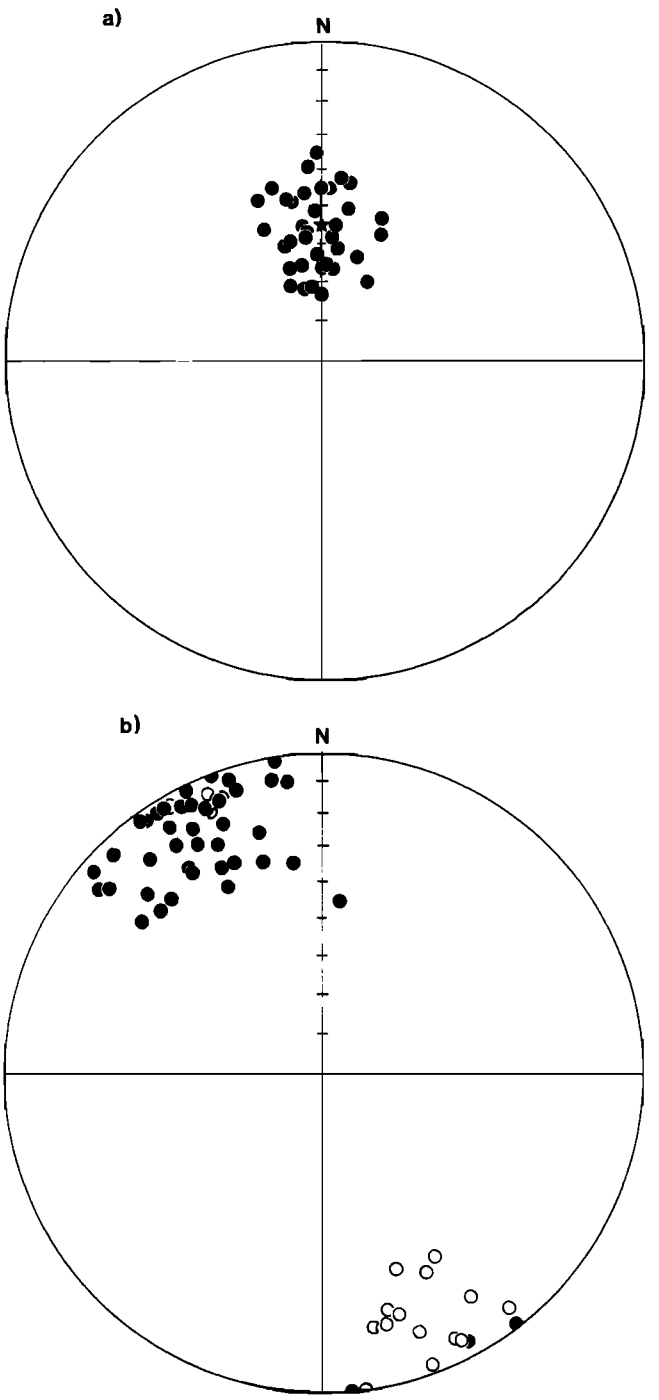


Fig. 4. Paleomagnetic directions plotted on equal area stereographic projections. (a) Component removed between 20° and $450^{\circ}C$; present dipole field is plotted as a star at inclination=55°. (b) Characteristic directions after structural correction.

sites. After removal of a northerly and steep component some samples from this site showed northerly and others southerly intermediate directions, without reaching stable endpoints. Three of the four samples collected at site 9 were destroyed during transportation or drilling. These sites were excluded from further analysis.

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TABLE 1. Paleomagnetic Data and Statistical Parameters

Site	N(n)/No.	J590	D, deg.	I, deg.	k	α 95	Tilt Corrected D, deg	Tilt Corrected I, deg.	Latitude (N)	Longitude (E)
2	3(3)/9	1.0	148.0	-26.4	55	16.7	148.8	-14.4	-49.7	313.0
3	6(9)/6	13.8	317.7	24.4	134	5.8	318.9	17.0	44.2	143.7
5	3(6)/5	9.3	327.5	0.1	75	14.3	327.5	0.1	43.8	126.7
6	5(11)/7	10.5	343.7	14.5	16	19.9	343.7	14.5	58.7	111.0
7	4(7)/5	8.2	323.6	8.8	76	10.6	323.4	6.1	43.4	133.4
8	5(8)/5	23.8	336.0	26.5	42	11.9	333.5	27.8	59.1	136.0
10	3(4)/4	4.0	342.5	-3.7	41	19.5	342.5	-3.7	50.2	106.9
11	6(10)/6	12.1	324.3	4.8	50	9.5	324.3	4.8	43.9	132.9
12	4(5)/5	12.5	323.1	7.8	68	11.2	323.1	7.8	44.2	135.5
13	4(7)/4	24.5	333.6	26.3	48	13.4	333.6	26.3	58.5	134.7
14	5(6)/6	10.5	151.3	-5.7	37	12.8	151.2	-5.7	-48.2	304.6
15	4(6)/5	1.9	168.9	-5.0	22	20.0	168.6	-1.8	-54.3	278.0
16	5(6)/5	8.4	163.2	-23.8	88	8.2	163.2	-23.8	-62.8	296.5
17	4(6)/4	18.2	338.9	3.1	77	10.5	338.9	3.1	51.2	113.4
18	4(9)/4	19.6	331.5	20.3	44	14.1	331.5	20.3	54.3	131.9
19	3(4)/3	11.0	148.2	-7.1	178	9.3	148.2	-7.1	-46.9	308.7
Mean					39.5	5.9	152.4	-11.1		

		Lat. (N)	Long. (E)	A 95	R	K
Mean Pole	Normal	50.7	128.3	5.7	10.85	65.7
	Reverse	53	121.2	10	4.94	59.3
	Combined	51.4	126.2	4.6	15.76	64.5

D, declination; I, inclination. N(n) is the number of samples (specimens) used in site mean calculation, No is the number of samples demagnetized. J590 is the average site intensity after treatment at 590°C.

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Dissolution and collapse structures occasionally occur in the underlying Alibates Formation, producing small-amplitude disturbances in beds of the overlying formations. Where measurable, bedding dips of the Dewey Lake do not exceed 5° except in two sites (2 and 3) at the lower Palo Duro Canyon locality. Although not statistically significant as a fold test, tilt corrected directions are more tightly grouped.

Directions of individual specimens were averaged to calculate mean sample directions and individual samples were combined to calculate site means. A minimum of three samples per site were used to calculate the site mean. Site means, statistical parameters, and virtual paleomagnetic poles are given in Table 1.

Interpretation

The actual mode and timing of remanence acquisition in red beds have been a matter of controversy [Elston and Purucker, 1979; Larson et al., 1982]. We consider the characteristic direction (Figure 4b) to represent that of the geomagnetic field in the Late Permian for the following reasons: (1) Both normal and reverse polarities are observed; furthermore they are antipodal. A reversal test, at the 95% level of confidence [McFadden and Lowes, 1981], indicates that we can not reject the hypothesis that the normal and reverse directions were drawn from the same population; thus they share a common mean. (2) The paleomagnetic pole calculated for this formation (at 126.2°E and 51.4°N; N=16; A95=5°) is different from any pole of younger age for North America, thus rendering the possibility of a later remagnetization extremely unlikely. (3) The observed polarity intervals, especially the normal polarity at the lower ash layer (Figure 5) are roughly layer parallel, although higher density sampling would provide further confirmation of this feature. The characteristic component is thus interpreted to be a "primary" chemical remanent magnetization (CRM) acquired during or rapidly after deposition.

Discussion of Magnetostratigraphic Implications

Stratigraphic sections of the three localities sampled are shown in Figure 5. Approximate positions of sampling sites are indicated as well as the ash layers observed by Kolker and Fracasso [1985]. Samples for each site were collected at discrete stratigraphic intervals within the sections, and in each site the characteristic direction shows only one polarity state. The site means show a simple magnetostratigraphy with reverse polarity at the base, normal polarity near the ash layer, reversed in the Upper Palo Duro section, and then normal again. The upper part of the sequence is uncertain, given that the density of sampling sites is not ideal because of the nature of exposures. The mixed polarity interval, bracketing the time of deposition of the ash layer, may correlate with the Illawara magneto-zone [Irving and Parry, 1963] and certainly occurs above the upper boundary of the Kiaman. The reversed directions observed at the bottom of the sequence could be interpreted as

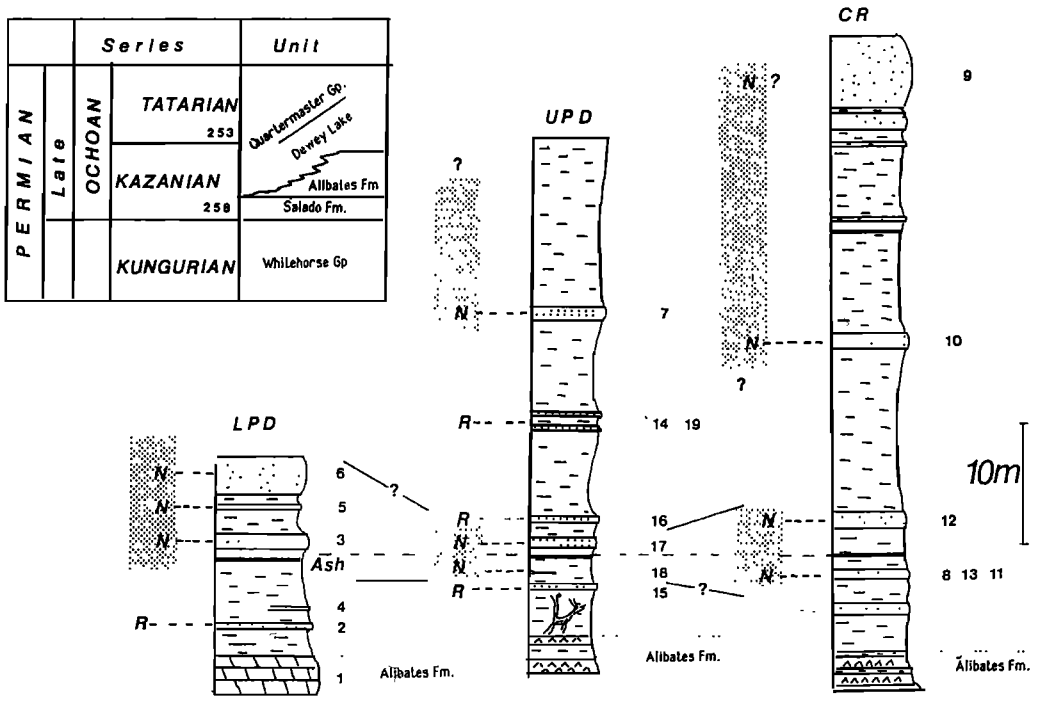


Figure 5. Schematic stratigraphic sections at the sampling localities, after Kolker and Fracasso [1985]. The ash layers are shown as solid lines, and the lower layer is correlated through the sections. The polarity obtained at each site is shown (N,R). Three polarity transitions define at least four polarity zones.

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the youngest rocks magnetized during the Kiaman Superchron, but this cannot be assessed because the polarity of the magnetization in the underlying Alibates Formation could not be determined.

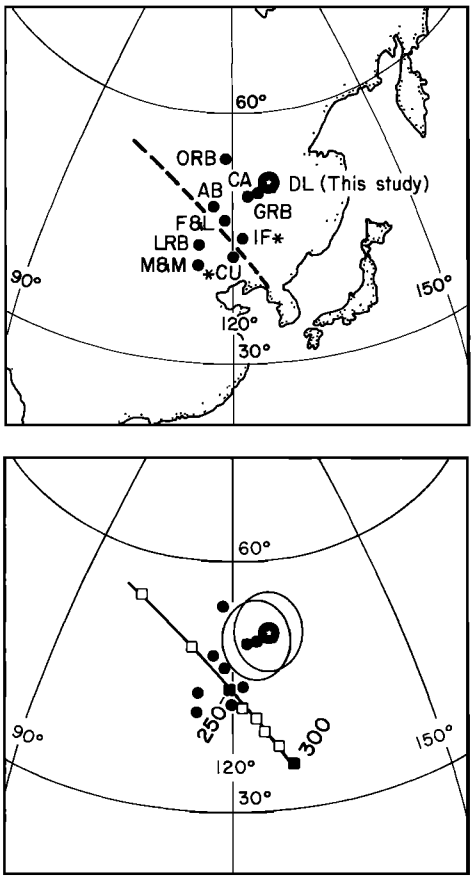


Figure 6. (a) Paleomagnetic pole from this study (DL) and other Early to Late Permian poles from cratonic North America. Confidence interval circles were excluded for clarity. The dashed line follows the trend of the PEP track of Gordon et al. [1984]. Early Permian paleomagnetic poles plotted are: CA, Casper Formation Wyoming [Diehl and Shive, 1981]; LRB, Leonardian red beds, SW United States [Peterson and Nairn, 1971]; M&M, Minturn and Maroon formations, Colorado [Miller and Opdyke, 1985]; F&L, Fountain and Lykins formations, Colorado [McMahon and Strangway, 1968]; CU, Cutler Formation, Colorado [Helsley, 1971]; IF, Ingleside Formation, Colorado [Diehl and Shive, 1979]; AB, Abo Formation, New Mexico [Steiner, 1988]. Late Permian poles plotted are: GRB, Guadalupian red beds, SW United States [Peterson and Nairn, 1971]; ORB, Ochoan red beds. SW United States [Peterson and Nairn, 1971]. An asterisk next to letters denotes poles from the Colorado plateau. (b) Same as Figure 6a, but including cones of confidence for the Late Permian poles. The Dewey Lake result supersedes the ORB result of Peterson and Nairn [1971] and the cone of confidence for this pole is not plotted. PEP poles of Gordon are plotted at 10-m.y. intervals for the period 300-230 Ma.

The upper boundary of the Kiaman in SW North America is known to occur within the Late Permian (Tatarian-Kazanian), as normal polarities were previously reported by Peterson and Nairn [1971] in at least three sites of three Late Permian formations (including one in the Dewey Lake Formation). This is in excellent agreement with results from the Soviet Union and China [Khramov and Rodinov, 1981; Steiner et al., 1989]. Although an older age for the end of the Kiaman can not be entirely excluded from consideration, the radiometric age of the ash layer indicates that the Kiaman Superchron ended at or before 251 ± 4 Ma.

North American APWP

When the paleomagnetic pole for the Dewey Lake Formation is compared with other Permian poles from stable North America and the PEP track of Gordon et al. [1984] (Figure 6), it is apparent that the Permian segment of the polar wander path exhibits a more north-south trend than the PEP track. In fact, the paleomagnetic pole trend is poorly aligned with the general NW-SE orientation of the PEP trend. The ability to fit small circles through APWP segments for a PEP determination implies that plate motions are constant with respect to the rotation axis for significant time intervals. When the PEP approach was first proposed there were few data for the Late Permian of North America and their reliability was not well established. The Dewey Lake pole, however, confirms the reliability of the Late Permian poles of Peterson and Nairn [1971]. These three Late Permian paleopoles suggest that plate motions during the Late Paleozoic-Early Mesozoic cannot be described by a single Euler rotation axis and that a more segmented set of small circles segments is needed to model the APWP. This, regrettably, diminishes the usefulness of the PEP method as it no longer provides accurate "synthetic" predictions of paleomagnetic pole positions useful for tectonic reconstructions. Recent studies by Bryan and Gordon [1986] and Steiner [1988] have discussed possible relative rotation of the Colorado Plateau with respect to North America. Even though some Permian poles in Figure 6 come from this area, our conclusions are the same with or without taking the rotation into account.

Acknowledgments. We wish to thank the Texas State Parks Department for their cooperation. We thank also JGR's reviewers, Steve May and Dennis Kent, for helpful comments. This study was partially supported by the New Mexico Bureau of Mines and Mineral Resources, special studios grant to J.W.G., the National Science Foundation Division of Earth Sciences, grant EAR 83-13085 to R.V., and a Sigma Xi Society grant to R.S.M-G.

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(Received February 24, 1989;
revised July 12, 1989;
accepted July 18, 1989.)

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